

EOM 231.02- Describe the production of lift by an aircraft wing

Importance

It is important for cadets to learn about the production of lift by an aircraft wing so that they can develop an understanding of subsequent and related principles of flight.

Overview

Teaching Points

TP 1. Explain that air acts like a fluid insofar as it has inertia, speed and pressure

TP 2. Introduce Bernoulli's principle which states that a liquids pressure drops when it's speed increases

TP 3. Have the cadets explore Bernoulli's principle

TP 4. Introduce angle of attack

TP5. Have each cadet create and fly an airfoil

TP 1. Explain that air acts like a fluid insofar as it has inertia, speed and pressure

Newton's laws of motion

First Law: An object in motion will remain in motion until acted upon by a net force

- Predicts that air being a gaseous fluid tends to remain in motion when it is moving

Second Law: Force is equal to mass times acceleration

- Requires that a force must be applied to change the air's motion

Third Law: A force will have an equal and opposite reaction force

- Allows the aircraft wing to develop lift by forcing air downward which allows a reaction force upwards on the wing

Confirmation Questions!

Q1. How does air follow Newton's first law of motion?

A1. When in motion, air tends to remain in motion.

Q2. How does air follow Newton's second law of motion?

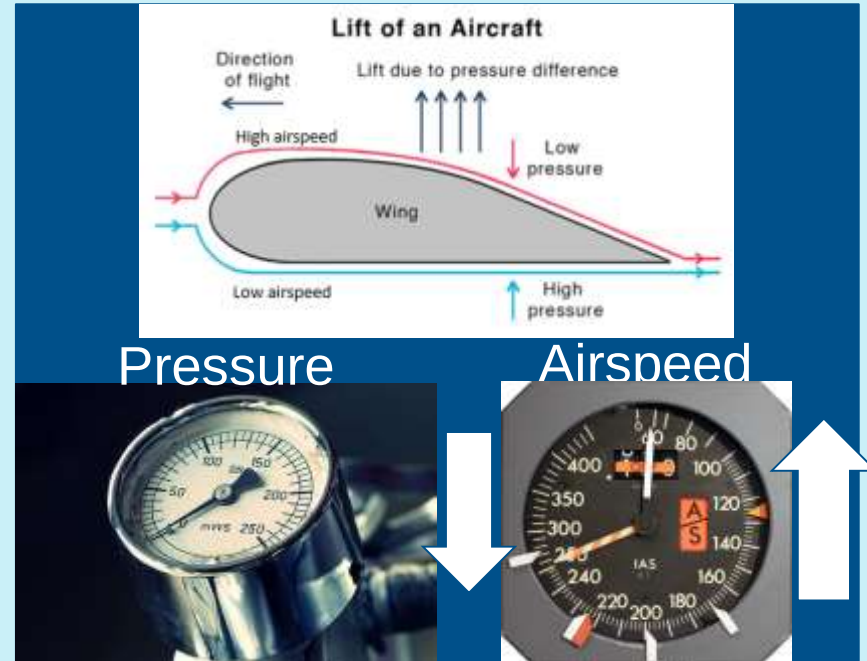
A2. When in motion, a force must be applied to change air's motion.

Q3. How does air follow Newton's third law of motion?

A3. When air's motion is changed, an equal and opposite reaction results.

TP 2. Introduce Bernoulli's Principle which states that liquid's pressure drops when it's speed increases

- To develop the equal and opposite reaction the wing is shaped to take advantage of Bernoulli's Principle (BP) to make air change direction
- Air behaves like a fluid because it has pressure and speed
 - As airspeed increases pressure drops
- A wing uses BP to deflect air which causes an equal and opposite reaction



Bernoulli's Principle

PAPER AIRFOIL ACTIVITY

- BERNOULLI'S PRINCIPLE STATES THAT THE TOTAL ENERGY IN A SYSTEM REMAINS CONSTANT.
- A FLUID SYSTEM HAS ENERGY IN THE FORM OF SPEED AND PRESSURE.
- IF SPEED INCREASES, PRESSURE DECREASES AND IF SPEED DECREASES, PRESSURE INCREASES.



Confirmation Questions!

Q1. What relationship exists between air speed and air pressure?

A1. An inverse relationship: as speed increases, pressure drops and as speed drops, pressure increases.

Q2. What is the wing's connection between Newton's third law and Bernoulli's Principle?

A2. A wing uses Bernoulli's Principle to deflect air, which causes an equal and opposite reaction.

Q3. What is a wing's shape calculated to do?

A3. A wing's shape is calculated to decrease pressure above while increasing pressure below.

TP3. Have the cadets explore Bernoulli's principle

Activity

The pressure of moving air can be examined by blowing gently over a small piece of curved paper. The air does not push the paper down as might be intuitively assumed. Instead, the paper behind the curve rises toward the moving air. This happens because the air pressure drops over the paper due to the air's increased speed – this would seem to match the description of speed/pressure relationship. The curvature in the paper enhances the effect of the lowered air pressure.

A similar effect can be observed when air moves past any object that is light enough to be affected by the drop in air pressure associated with movement. A balloon is light enough to show this effect clearly.

Creating lift activity



Confirmation Questions!

Q1. Why does the paper airfoil rise when a cadet blows over it?

A1. Air pressure over the paper drops as the air moves, so the still air below the paper pushes it up.

Q2. Why do balloons move toward moving air?

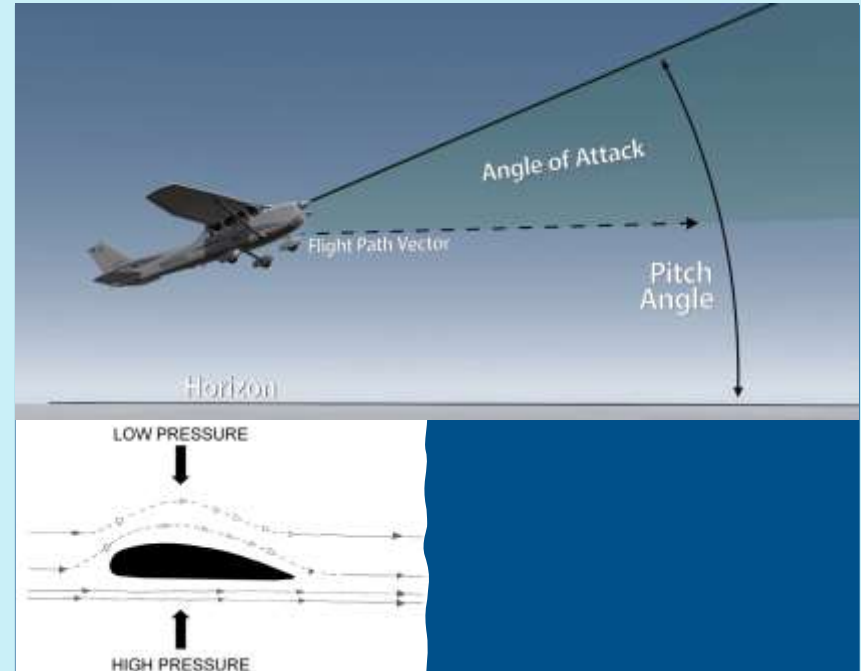
A2. Moving air pressure is lower, so still air pushes the balloon into the moving air.

Q3. Why was the paper deliberately curved before blowing over it?

A3. The curvature in the paper enhances the effect of the lowered air pressure.

TP 4. Introduce angle of attack

- An aircraft wing is an airfoil because of its cross-sectional shape
- The top surface is curved outward (convex) allowing air to flow overtop further, making the air travel faster which results in lower pressure above the wing
- Below the wing air is slowed to increase its pressure done by curving the surface slightly inward (concave) and by sloping the wing so that it's slightly higher at the front (leading edge) than the back (trailing edge)
- This angle of the wing's under surface which encounters moving air is called the wing's angle of attack



Confirmation Questions!

Q1. Which surface of an aircraft wing is curved outward (convex)?

A1. The top surface of a wing is curved outward.

Q2. What shape is often given to the underside of an aircraft wing?

A2. A slight inward, or concave, curve is often given to an aircraft wing.

Q3. What happens to air pressure under a wing as angle of attack increases?

A3. When air's motion is changed, an equal and opposite reaction results.

TP 5. Have each cadet create and fly an airfoil

Activity

ACTIVITY INSTRUCTIONS

Give each cadet an index card to fold into an airfoil shape with mounting holes for the launching skewers. The cadets will:

1. Fold the card in half. Tape the top of the card down to the bottom half of the card so that about 6 mm (1/4 inch) of the bottom shows, leaving the top in a curve and the bottom flat. This will produce a cross-section airfoil shape.
2. Use the hole punch (or a sharp pencil) to put two sets of holes in the thickest part of the airfoil.
3. Cut the straw so you get two pieces 5 cm (2 inches) long. Fit these mini-straws through the holes in the airfoil.
4. Set the airfoil on the base and slip a skewer or unbent paper clip through each of the mini-straws and into the base.
5. Use the hairdryer or fan to move air over the airfoil to create lift. Try it both ways—with the flat side of the airfoil on the bottom and with the flat side on the top. Note which position the airfoil is in when it climbs best.

AN AIRFOIL SECTION



End of Lesson Confirmation

Q1. How does the top surface of a wing generate lift?

A1. The top surface of a wing generates lift by lowering air pressure over the wing.

Q2. How does the under-surface of a wing generate lift?

A2. The under-surface of a wing generates lift by raising air pressure under the wing.

Q3. What determines how much lift is produced by a wing at a given speed?

A3. The angle of attack will determine how much lift is produced at a given speed.

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